FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Cerebral Palsy

Tutorial | Week 4 · Small babies, sugar & the heart | Paediatrics | Dr Nabatte

Cerebral palsy is a permanent, non-progressive disorder of movement and posture caused by an injury or malformation of the developing brain (before birth to about 2 years); it is common in Uganda (about 3 per 1000 children), where birth asphyxia, kernicterus, prematurity, cerebral malaria and meningitis are leading causes, and good care is multidisciplinary, family-centred and focused on function, nutrition, seizures, hips and communication.

Must know before the ward round

1. CP = permanent, NON-progressive brain lesion (fetal life to ~2 years) → disorder of movement and posture; the clinical picture changes with growth but the lesion does not. Loss of skills = NOT CP: look for a progressive disease.
2. Motor types: spastic ~70–80% (UMN signs), dyskinetic 10–15% (dystonic/choreoathetoid; basal ganglia: HIE, kernicterus), ataxic <5% (cerebellar), mixed.
3. Topography: hemiplegia (unilateral; perinatal MCA stroke), diplegia (legs > arms; preterm periventricular leukomalacia), quadriplegia (all four limbs, often with epilepsy and severe ID; term HIE, meningitis, cerebral malaria).
4. GMFCS (age 6–12): I walks without limits; II walks with limits; III walks with a hand-held mobility device; IV self-mobility limited (powered mobility); V transported in a manual wheelchair. Stable after age 2; predicts hips, nutrition and survival.
5. Early warning signs (NICE NG62): not sitting by 8 months, not walking by 18 months, hand preference before 1 year, persistent fisting, toe-walking, feeding difficulty, abnormal tone; persistent primitive reflexes beyond 6 months.
6. Uganda: prevalence ~3/1000 (Iganga-Mayuge study, 2017); causes: birth asphyxia/HIE, prematurity, neonatal jaundice (kernicterus), neonatal sepsis/meningitis, cerebral malaria, bacterial meningitis, head injury. Test every child with developmental delay for HIV (UCG 2023).
7. Comorbidities: epilepsy 15–60% (~1 in 3), intellectual disability 30–50%, vision (strabismus, cortical visual impairment), hearing loss (kernicterus), feeding/swallowing problems → malnutrition and aspiration, drooling, GORD, constipation, hip displacement, scoliosis, osteopenia, pain, sleep and behaviour problems.
8. Leading causes of death: aspiration pneumonia and malnutrition. Weigh, measure (or MUAC) and assess swallowing at every visit.
9. Generalised spasticity: oral baclofen 0.3 mg/kg/day in 3–4 divided doses, increase every 3 days to 0.75–2 mg/kg/day (max 40 mg/day <8 y, 60 mg/day ≥8 y); never stop suddenly (withdrawal seizures, hallucinations). Diazepam 0.1–0.8 mg/kg/day divided 6–8-hourly (short-term, spasms, night-time).
10. Focal spasticity: botulinum toxin A + casting/orthoses (AFOs) + physiotherapy; dystonia: trihexyphenidyl (specialist); always try levodopa if dystonia is unexplained (dopa-responsive dystonia mimics CP).
11. Epilepsy in CP (UCG 2023): <2 y phenobarbital or carbamazepine; >2 y carbamazepine (5 → 10–20 mg/kg/day) or valproate (10 → 20–30 mg/kg/day); avoid phenobarbital and phenytoin if intellectual disability or behaviour problems.
12. Hip surveillance: hip X-ray (migration percentage) from ~2 years, especially GMFCS III–V; refer to orthopaedics if migration >30–40% or painful/limited abduction.

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In a nutshell

Cerebral palsy (CP) is a group of permanent disorders of **movement and posture** that limit activity, caused by a **non-progressive** disturbance in the **developing brain** (in the womb, at birth or in early childhood, up to about 2 years). The motor problem is often accompanied by **epilepsy, intellectual disability, problems with vision, hearing, communication, feeding and behaviour**, and later by secondary **musculoskeletal** problems (contractures, hip dislocation, scoliosis).

It is the **commonest cause of physical disability in children**. In Uganda about **3 in every 1000 children** have CP, and many more of them are severely affected, malnourished and without access to rehabilitation than in high-income countries. A large share of Ugandan CP is **preventable**: good intrapartum care, neonatal resuscitation, treating neonatal jaundice, preventing and treating malaria and meningitis, and immunisation.

> CLINICAL PEARL
> The one thing never to forget: **CP does not get worse.** The child gains skills slowly, but never loses them. A child who **loses** skills (regression), has a progressive course, or has no risk factor in the history does not have CP until a progressive or treatable disorder (HIV encephalopathy, metabolic, neurodegenerative, dopa-responsive dystonia, a spinal or brain tumour) has been excluded.

Definitions & classification

Definition (Rosenbaum et al., 2007, still the accepted definition): "Cerebral palsy describes a group of permanent disorders of the development of movement and posture, causing activity limitation, that are attributed to non-progressive disturbances that occurred in the developing fetal or infant brain. The motor disorders of CP are often accompanied by disturbances of sensation, perception, cognition, communication and behaviour, by epilepsy, and by secondary musculoskeletal problems."

Four things classify a child with CP:

1. Motor type (the main tone abnormality)

Type	Proportion	Lesion	Clinical features
Spastic	~70–80%	Upper motor neurone: motor cortex, periventricular white matter (corticospinal tracts)	Velocity-dependent increased tone ("clasp-knife"), brisk reflexes, clonus, upgoing plantars, persistent primitive reflexes, contractures, scissoring, toe-walking

Type	Proportion	Lesion	Clinical features
Dyskinetic (dystonic, choreoathetoid)	~10–15%	Basal ganglia and thalamus: term HIE (acute profound asphyxia), **kernicterus**	Fluctuating tone, involuntary writhing or twisting movements and postures worse with effort or emotion, dysarthria, drooling; intelligence often better than it looks; kernicterus adds **sensorineural deafness, upward-gaze palsy and dental enamel dysplasia**
Ataxic	<5%	Cerebellum	Hypotonia, wide-based unsteady gait, intention tremor, dysmetria, nystagmus. **Look hard for a genetic or progressive cause**
Mixed	Common	Several areas	Usually spastic–dystonic

NOTE

Hypotonic infants may later declare themselves as spastic or dyskinetic: many children with CP are floppy in the first months and become stiff later. Truly persistent hypotonia without other signs is not usually called CP; think of genetic, neuromuscular or metabolic causes.

2. Topography (which limbs)

Pattern	Description	Typical cause
Hemiplegia (unilateral spastic)	One side; arm usually worse than leg	Perinatal arterial stroke (MCA territory), unilateral malformation, postnatal stroke (**sickle cell disease**), head injury
Diplegia (bilateral, legs worse)	Legs much more affected than arms; scissoring, toe-walking; IQ often normal	**Preterm** birth with **periventricular leukomalacia** (PVL) or intraventricular haemorrhage (leg fibres run closest to the ventricles)
Quadriplegia (bilateral, all four limbs, also called tetraplegia)	All four limbs, trunk, bulbar muscles; often epilepsy, severe intellectual disability, microcephaly, feeding problems	Term **HIE** (watershed injury), **meningitis**, **cerebral malaria**, severe hypoglycaemia, congenital infection (CMV), brain malformation

The European SCPE system simply uses **unilateral vs bilateral spastic**, dyskinetic and ataxic.

3. Function: the Gross Motor Function Classification System (GMFCS, expanded and revised 2007)

Level	Between 6th and 12th birthday	Prognosis / priorities
I	Walks indoors and outdoors and climbs stairs **without limitations**; runs and jumps but with reduced speed, balance and coordination	Near-normal mobility
II	**Walks with limitations**: difficulty on long distances, uneven ground, crowds; uses a rail on stairs; minimal running or jumping	Independent walker with limits
III	**Walks using a hand-held mobility device** (walker, crutches) indoors; wheelchair for longer distances	Hip surveillance; orthoses
IV	**Self-mobility with limitations**; may use **powered mobility**; transported in a manual wheelchair outdoors	High risk: hip dislocation, scoliosis, feeding problems
V	**Transported in a manual wheelchair**; limited head and trunk control; all areas of motor function limited	Highest risk: aspiration, malnutrition, hip dislocation, early death

Companion systems: **MACS** (Manual Ability Classification System) for hand use, **CFCS** (Communication Function Classification System) and **EDACS** (eating and drinking).

EXAM FAVOURITE

Prognosis for walking: a child who **sits independently by 2 years** will very likely walk (with or without aids); a child who cannot sit by about 4 years is unlikely ever to walk. GMFCS level is fairly stable after about age 2, so it is the best single predictor of future mobility, hip displacement and need for nutritional support.

4. Timing and cause

Timing	Share (HIC)	Examples
Prenatal	~70–80% in HICs (lower in Uganda)	Brain malformations, congenital infection (CMV, rubella, toxoplasmosis, Zika), maternal infection/chorioamnionitis, placental insufficiency and fetal growth restriction, multiple pregnancy, fetal stroke, genetic causes, maternal thyroid disease, toxins
Perinatal	~10–20%	**Birth asphyxia/HIE**, prematurity complications (PVL, IVH), **neonatal sepsis/meningitis**, **hypoglycaemia**, **kernicterus**, birth trauma
Postnatal (to ~2 years)	~10% in HICs, **much higher in Uganda**	**Cerebral malaria**, **bacterial meningitis**, encephalitis, **head injury**, near-drowning, **sickle cell stroke**, severe dehydration with venous sinus thrombosis, poisoning

EXAM FAVOURITE

When can CP be blamed on birth asphyxia? Only with evidence of **intrapartum hypoxia–ischaemia** (severe metabolic acidosis in cord or early neonatal blood, low Apgar scores), **moderate or severe neonatal encephalopathy** soon after birth, a **spastic quadriplegic or dyskinetic** pattern, and **exclusion of other causes**. Most CP in HICs is not caused by events in labour; in Uganda birth asphyxia is a much bigger contributor.

Epidemiology & Uganda context

- **Global:** about **2–3.5 per 1000 live births** in high-income countries, falling slightly as neonatal care improves; much higher in preterm and very-low-birth-weight babies (risk rises steeply below 32 weeks).
- **Uganda:** a population-based study in the **Iganga-Mayuge** Health and Demographic Surveillance Site (Kakooza-Mwesige et al., *Lancet Global Health* 2017) found a prevalence of roughly **3 per 1000 children** aged 2–17 years. Compared with HIC registers, Ugandan children were **older at diagnosis**, more **severely affected** (more GMFCS IV–V), more often **malnourished**, had more **postnatal causes** (malaria, meningitis, other infections) and more **epilepsy**; few had ever received rehabilitation or assistive devices.
- **Preventable causes in Uganda:** birth asphyxia (obstructed and prolonged labour, poor monitoring), **severe neonatal jaundice** (kernicterus; G6PD deficiency, ABO incompatibility, home births, delayed phototherapy), neonatal sepsis and meningitis, hypoglycaemia, **cerebral malaria**, **bacterial meningitis** (Hib and pneumococcus, now reduced by vaccines), **measles encephalitis**, **sickle cell stroke**, head injury.
- **Mortality:** children with severe CP (GMFCS IV–V), feeding difficulties and malnutrition die young, mostly from **aspiration pneumonia** and malnutrition. In Ugandan cohorts, mortality is many times higher than in children without CP.
- **Stigma:** CP is often blamed on witchcraft, curses or the mother. Children may be hidden at home, mothers abandoned by partners, and children excluded from school. Families spend heavily on traditional healers before reaching a rehabilitation service.

UGANDA CONTEXT

Rehabilitation services exist but are concentrated in and around Kampala: **CoRSU Rehabilitation Hospital (Kisubi)**, **Mulago** physiotherapy/occupational therapy and paediatric neurology clinics, **Katalemwa Cheshire Home**, **Mengo** physiotherapy, regional referral hospital physiotherapy departments and **community-based rehabilitation (CBR)** programmes run by NGOs and churches. Group-based parent training programmes developed and tested in Uganda (e.g. **"Getting to Know Cerebral Palsy"** and the **ABAaNA** early-intervention programme) teach parents positioning, feeding, play and communication and reduce stigma. Botulinum toxin, intrathecal baclofen, selective dorsal rhizotomy and MRI are largely unavailable or very expensive.

Pathophysiology

1. **Lesion location determines the type.** The developing brain's vulnerability changes with gestational age:
   - **24–34 weeks:** the **periventricular white matter** is a vascular watershed and its oligodendrocyte precursors are very sensitive to ischaemia and inflammation → **PVL** → damage to corticospinal fibres for the **legs** (which run closest to the ventricles) → **spastic diplegia**. IVH with venous infarction → hemiplegia.
   - **Term:** grey matter is most active. **Acute profound asphyxia** (cord prolapse, uterine rupture, abruption) damages the metabolically active **basal ganglia and thalamus** → **dyskinetic** CP. **Prolonged partial asphyxia** injures **parasagittal cortex and watershed zones** → **spastic quadriplegia**. Perinatal **arterial stroke** (usually left MCA) → **hemiplegia**.
   - **Unconjugated bilirubin** crosses an immature or damaged blood–brain barrier and deposits in the **globus pallidus, subthalamic nucleus, hippocampus, brainstem auditory nuclei and oculomotor nuclei** → **kernicterus**: dyskinetic CP, sensorineural hearing loss, upward-gaze palsy, dental enamel dysplasia.
   - **Postnatal infection** (cerebral malaria, meningitis): diffuse cortical injury, infarcts, hydrocephalus → quadriplegia, hemiplegia, epilepsy, deafness, blindness.
2. **Why spasticity develops:** loss of descending inhibition from the cortex onto spinal reflex arcs → **hyperexcitable stretch reflexes**, co-contraction of agonists and antagonists, persistent primitive reflexes, and weakness and poor selective motor control.
3. **Why the picture "changes" although the lesion does not:** as the child grows, **muscles fail to grow as fast as bones** (spastic muscle is short and stiff) → **contractures**; abnormal muscle pull on growing bones → **hip migration**, **scoliosis**, foot deformities (equinus, planovalgus). Tone often appears first as **hypotonia** in early infancy, with spasticity or dystonia becoming obvious at 6–18 months as myelination proceeds.
4. **Why nutrition fails:** oromotor dysfunction (poor lip closure, tongue thrust, uncoordinated swallow), long feeding times, GORD, constipation, aspiration, and high energy needs in dyskinetic CP (low needs in severe spastic quadriplegia) → **undernutrition, osteopenia, poor immunity, pressure sores**.

Clinical acumen

Focused history

- **Presenting concern**: delayed motor milestones, stiffness or floppiness, abnormal posture, early hand preference, toe-walking, feeding problems, fits, "not like other children".
- **Developmental history by domain** (gross motor, fine motor/vision, language/hearing, social): age at head control, sitting, crawling, standing, walking; first words; smile. **Has the child ever lost a skill?** (regression = not CP).

- **Antenatal**: maternal illness, fever, rash (rubella, CMV), HIV and syphilis status, bleeding, pre-eclampsia, multiple pregnancy, drugs/alcohol, fetal movements.
- **Birth**: gestation, place of birth (home/HC/hospital), duration of labour, obstructed labour, cord prolapse, mode of delivery, **cried at birth?**, **resuscitation**, Apgar, birthweight, **fits in the first days**, admission to the neonatal unit, **jaundice** (when, how deep, phototherapy, exchange transfusion), hypoglycaemia, sepsis/meningitis.
- **Postnatal**: cerebral malaria, meningitis, convulsions, head injury, near-drowning, severe dehydration, measles, **sickle cell disease**.
- **Associated problems**: seizures (type, frequency, drugs, adherence), vision (does she follow faces?), hearing (turns to sound?), communication, **feeding** (time per meal, coughing/choking during feeds, chest infections, vomiting), drooling, constipation, sleep, pain (hips, constipation, dental), behaviour, continence, school attendance.
- **Nutrition and feeding**: breastfeeding history, current diet (thin porridge only?), texture, who feeds the child and how.
- **Immunisation** (UNEPI card) and **HIV status** of child and mother.
- **Family**: consanguinity, similar conditions (genetic), siblings' health.
- **Social**: who cares for the child, distance to services, cost, support from partner, stigma, schooling, previous treatments (traditional healers, physiotherapy, devices).

Examination

- **Growth**: weight, length/height (or knee height/ulna length if contractures prevent standing), **head circumference** (microcephaly is common in quadriplegia; macrocephaly suggests hydrocephalus), **MUAC**, Z-scores, oedema. Plot over time.
- **General**: dysmorphic features (syndromic cause), neurocutaneous signs, skin (pressure sores), teeth (caries, enamel dysplasia in kernicterus), chest (aspiration pneumonia), signs of HIV.
- **Observation** before touching: posture (frog-leg hypotonia; scissoring; fisting with adducted thumb; opisthotonus; asymmetry), spontaneous movements, involuntary movements, how the child sits, crawls or walks, hand use.
- **Tone**: pull-to-sit (head lag), ventral suspension (floppy "inverted U"), **vertical suspension** (scissoring legs in spastic CP; slips through hands if hypotonic), passive movement of limbs (spastic catch, "clasp-knife"; lead-pipe rigidity in dystonia), **Modified Ashworth** score in older children.
- **Power and selective control** (functional in young children); **reflexes** (brisk, spreading, **clonus**, upgoing plantar after ~12–18 months); **primitive reflexes** (Moro, ATNR, palmar grasp persisting beyond 4–6 months); **protective reflexes** (parachute absent or asymmetric after 9 months).
- **Musculoskeletal**: range of hip abduction (limited = hip at risk), Thomas test (hip flexion contracture), popliteal angle (hamstrings), ankle dorsiflexion (equinus), leg-length discrepancy, **spine** (scoliosis), feet.
- **Gait**: toe-walking, scissoring (diplegia), circumduction with arm flexed (hemiplegia), wide-based (ataxia), dystonic posturing.
- **Vision** (fix and follow, squint, nystagmus), **hearing** (response to sound, otoscopy), **communication**, **oromotor** (drooling, lip closure, observe a feed if possible).
- **Classify** at the end: motor type, topography, GMFCS level, associated impairments.

Red flags

RED FLAG

Refer for paediatric neurology and look for another diagnosis (NICE NG62) if:

- **No known risk factor** in the history
- **Loss of skills** (developmental regression) or a progressive course
- **Family history** of a progressive neurological disorder
- **New or unexpected focal neurological signs**
- Signs of **MRI-negative** dystonia or fluctuating symptoms (try **levodopa**: dopa-responsive dystonia)
- Abnormal head growth (crossing centiles upward = hydrocephalus/tumour; acquired microcephaly with regression = HIV encephalopathy, Rett syndrome)
- Peripheral signs: **absent reflexes**, wasting, fasciculation, sensory level, bladder/bowel problems (spinal cord or neuromuscular disease)

Thinking like a clinician

- **"Delayed walking at 18 months in a boy who walks on his toes and has big calves"**: think **Duchenne muscular dystrophy** (Gowers sign, raised CK), not CP.
- **"Floppy baby who later becomes stiff"** is typical of evolving CP; **"floppy and weak but alert"** suggests a neuromuscular cause (spinal muscular atrophy: absent reflexes, tongue fasciculation).
- **"Dystonia that is worse in the evening and better after sleep"**: dopa-responsive dystonia; a levodopa trial transforms the child.
- **HIV encephalopathy** presents with delayed or **lost** milestones, **acquired microcephaly** and spastic diplegia/quadriplegia. UCG 2023: **test every child with developmental delay for HIV** and start ART.
- **Kernicterus history** (deep jaundice in week 1, exchange transfusion, "the baby's eyes were yellow and he stopped breastfeeding") + dyskinetic CP + deafness: arrange a **hearing test** early.
- **Poor weight gain** in CP is not "just CP": it is a treatable feeding problem (and sometimes TB, HIV or neglect).
- **Pain** is often missed: the irritable child with CP may have a **dislocated hip, constipation, dental abscess, reflux oesophagitis, fracture (osteopenia)** or urinary infection.
- **A child who "refuses to eat" and coughs at meals** is aspirating; this is the commonest route to death in severe CP.
- **A preterm survivor with legs stiffer than arms** = spastic diplegia from PVL; arms are involved too if you look carefully (fine motor clumsiness).

Clinical workup

Step 0: First 5 minutes

Most children with CP come to clinic, not to the emergency room. But on the ward they often arrive **acutely ill**:

1. **ETAT**: airway (secretions, poor head control, aspiration), breathing (pneumonia, aspiration), circulation (dehydration from poor intake), **convulsions** (status epilepticus: treat as in the febrile convulsions topic), **glucose** (malnourished children are often hypoglycaemic).
2. **Position safely**: side-lying or supported upright, suction, oxygen if SpO_2 <90% (or <94% if in distress), nil by mouth if drowsy or choking.
3. **Nutrition emergency**: MUAC, oedema, WHZ → SAM protocol if indicated.
4. **Pain or a new deformity** after minor handling: think **fracture** (osteopenia) and splint.

Step 1: Focused history

- [ ] **Milestones and regression** → confirms delay; any loss points away from CP.
- [ ] **Antenatal, birth, neonatal and postnatal events** → cause and timing (and whether it was preventable, which matters for counselling and MPDSR-type learning).
- [ ] **Seizures and current drugs** → epilepsy control; drug side-effects (sedation, drooling).
- [ ] **Feeding** (time, coughing, chest infections, texture) → aspiration risk, need for SLT/NG/gastrostomy.
- [ ] **Vision and hearing** → early referral.
- [ ] **Pain, sleep, constipation, behaviour** → treatable problems.
- [ ] **HIV status, immunisation** → HIV encephalopathy; preventable infections.
- [ ] **Family and social situation** → caregiver capacity, stigma, school, financial support.

Step 2: Focused examination

Growth (Z-scores, HC, MUAC), dysmorphism, neurocutaneous signs, tone, power, reflexes, primitive and protective reflexes, involuntary movements, hips (abduction), spine, contractures, gait, vision, hearing, oromotor function, chest, teeth, skin. Then classify (type, topography, GMFCS).

Document these positives	And these negatives
Type, topography, GMFCS level	"No regression; no loss of skills"
Tone (Ashworth), reflexes, clonus, contractures	"Head circumference following centile"
Hip abduction range, spine	"No neurocutaneous stigmata, no dysmorphism"
Weight, height, MUAC, Z-scores	"HIV test negative; no chest signs"

Step 3: Bedside tests

Test	How	What it tells you
Weight, length/height, HC, MUAC → Z-scores	Scales, length board, tape; knee height if contractures	Malnutrition, microcephaly, growth faltering
Observed feed	Watch the carer feed thin and thick textures, child upright	Coughing, wet voice, desaturation = aspiration risk
SpO_2	Pulse oximeter, also during feeding	Aspiration, pneumonia
Glucose	Glucometer	Hypoglycaemia in ill, malnourished children
HIV rapid test (DNA PCR <18 months if exposed)	Per national algorithm	HIV encephalopathy
Simple hearing and vision screen	Response to voice/rattle; fix and follow, red reflex	Refer for formal testing
Developmental screening	Milestone checklist; Malawi Developmental Assessment Tool (MDAT) or similar	Global vs motor-only delay

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management
Cranial ultrasound	Neonates and infants with an open fontanelle at risk (preterm, HIE)	PVL (cysts), IVH, hydrocephalus	Early diagnosis, follow-up. Available at most hospitals with a skilled sonographer

Test	When	Expected / abnormal finding	How it changes management
MRI brain	Every child with suspected CP **if available** (abnormal in ~80–90%)	PVL, basal ganglia/thalamic injury, infarct, malformation, kernicterus (globus pallidus), normal (think genetic/metabolic)	Confirms timing and cause; normal MRI = look for another diagnosis. **Mulago, Mengo and private centres in Kampala; costly**
CT brain	When MRI not available	Calcification (CMV), infarcts, hydrocephalus, atrophy	Cause; hydrocephalus needs neurosurgery. Mulago, Mengo, some RRHs
EEG	Seizures of uncertain type, suspected epileptic syndrome	Epileptiform activity, hypsarrhythmia (infantile spasms)	Antiepileptic choice. Mulago, CoRSU, private
HIV test, FBC, sickle test	All with developmental delay (HIV per UCG); sickle test if hemiplegia	HIV, HbSS	ART; stroke prevention (hydroxyurea, transfusion programme)
TSH/T4, CK, metabolic and genetic tests	Hypotonia, ataxia, regression, normal MRI, family history, dysmorphism	Hypothyroidism; raised CK (Duchenne); specific metabolic disease	Treatable mimics. Referral labs; genetic testing mostly abroad/research
TORCH serology / CMV PCR	Microcephaly, calcification, deafness, chorioretinitis	Congenital infection	Counselling, hearing follow-up
Hip X-ray (AP pelvis)	From about 2 years, especially GMFCS III–V; repeat 6–12-monthly	**Reimers migration percentage**: >30% = hip at risk; >40% = displacement	Orthopaedic referral (soft-tissue release, osteotomy). Hospitals
Spine X-ray	Clinical scoliosis	Cobb angle	Bracing/surgery referral
Hearing test (OAE/ABR) and ophthalmology	All children with CP early on	Sensorineural loss (kernicterus, meningitis, CMV); cortical visual impairment, strabismus, refractive error	Hearing aids, glasses, communication plan. Mulago ENT and eye clinics, CoRSU
Chest X-ray	Recurrent chest infections	Right-lower-lobe consolidation (aspiration)	Feeding plan, antibiotics

CLINICAL PEARL

Early diagnosis is now possible before 6 months corrected age in high-risk infants, using **MRI** plus the **Prechtl General Movements Assessment** (absent "fidgety" movements at 3–5 months corrected age is highly predictive) or the **Hammersmith Infant Neurological Examination (HINE)**. Even where these are not available, a high-risk baby (asphyxia, preterm, kernicterus, meningitis) should get **enhanced developmental follow-up** to 2 years (NICE NG62), because early intervention while the brain is most plastic improves function.

Step 5: Putting it together

Model one-line summary

> "A 4-year-old boy with spastic quadriplegic cerebral palsy, GMFCS level V, following severe birth asphyxia, complicated by epilepsy on carbamazepine, severe feeding difficulty with probable chronic aspiration, moderate underweight and stunting, bilateral hip adduction contractures and a right hip at risk, now

> admitted with aspiration pneumonia."

Problem list (always organise CP by problems, not just the diagnosis)

1. Acute: aspiration pneumonia, hypoxaemia.
2. Motor: spastic quadriplegia, GMFCS V, adductor and hamstring contractures, right hip migration.
3. Epilepsy: monthly seizures on carbamazepine (check dose/adherence).
4. Nutrition: WAZ −2.6, HAZ −1.8, oromotor dysfunction, thin-porridge diet.
5. Communication, vision, hearing: to be assessed.
6. Family: single mother, stigma, far from services.

EXAM FAVOURITE

Worked example: hip surveillance X-ray. A 5-year-old with GMFCS IV CP has painful nappy changes. AP pelvis: on the right, 45% of the femoral head lies lateral to Perkin's line (**migration percentage 45%**); on the left 20%. **Interpretation:** right hip displacement (>40%), left within normal limits. **Action:** orthopaedic referral (adductor release ± femoral/pelvic osteotomy), pain relief, positioning (abduction wedge), repeat the left hip X-ray in 6–12 months.

Differential diagnosis

Condition	Distinguishing features	Key investigation
HIV encephalopathy	Regression or plateau, acquired microcephaly, spasticity, other HIV features	HIV test; CD4/viral load
Neurodegenerative / metabolic disease (leukodystrophies, mitochondrial, lysosomal)	Regression, progressive course, consanguinity, organomegaly, seizures, vision loss	MRI, metabolic screen, enzyme/genetic tests
Duchenne muscular dystrophy	Boy, toe-walking, Gowers sign, calf pseudohypertrophy, normal early milestones then weakness	**CK markedly raised**
Spinal muscular atrophy / other neuromuscular	Floppy, weak, **absent reflexes**, tongue fasciculation, alert face	Genetic test, EMG
Dopa-responsive dystonia	Dystonia (often legs), diurnal variation, normal MRI	**Levodopa trial**
Hereditary spastic paraplegia	Slowly progressive spastic legs, family history	Genetics
Spinal cord lesion (spina bifida, tethered cord, TB spine, tumour)	Leg weakness, sensory level, bladder/bowel problems, back findings	MRI spine
Hydrocephalus / brain tumour	Large or growing head, vomiting, headache, papilloedema, regression	Cranial USS/CT/MRI
Rett syndrome	Girl, normal to 6–18 months then regression, hand-wringing, acquired microcephaly	*MECP2* testing
Congenital hypothyroidism	Hypotonia, large tongue, constipation, prolonged jaundice, delay	TSH, T4
Global developmental delay without CP / genetic syndromes (Down syndrome)	Hypotonia, dysmorphism, delay in all domains without UMN signs	Karyotype, genetic tests
Severe malnutrition / rickets	Delay that improves with feeding; bony signs of rickets	Growth, X-ray wrist, ALP

Investigations

Test	What you are looking for	Why
Growth and MUAC	Malnutrition	Commonest treatable problem; linked to death
HIV test	HIV encephalopathy	Treatable; UCG requires it in delay
Neuroimaging (USS, CT, MRI)	Cause, timing; normal MRI → other diagnosis	Diagnosis and counselling
EEG	Epilepsy type	Drug choice
Hearing and vision assessment	Sensory impairments	Communication, learning
Hip and spine X-rays	Migration, scoliosis	Prevent painful dislocation
CK, TSH, metabolic/genetic	Mimics	Avoid missing treatable or inherited disorders

Management

The aims are not to "cure the brain" but to **maximise function and participation, prevent secondary complications, keep the child well nourished and pain-free, control seizures, and support the family**. Care is **multidisciplinary, family-centred and lifelong**, and goals are set with the family (e.g. "sit to eat", "walk to school with crutches", "communicate yes/no").

A. The multidisciplinary team

Team member	Role
Paediatrician / paediatric neurologist	Diagnosis, coordination, epilepsy, spasticity drugs, nutrition, general health
Physiotherapist	Positioning, stretching, strengthening, mobility training, standing frames, walking aids, seating; teaches the family a **home programme**
Occupational therapist	Hand function, daily living skills (feeding, dressing), adaptive seating and equipment, splints
Speech and language therapist	**Swallowing safety**, feeding techniques, communication (including augmentative and alternative communication: picture boards, signs)
Nutritionist / dietitian	Energy and texture plan, NG or gastrostomy feeding, micronutrients
Orthopaedic surgeon and orthotist	AFOs (ankle–foot orthoses), serial casting, hip surveillance and surgery, scoliosis, tendon lengthening
Neurosurgeon	Hydrocephalus; selective dorsal rhizotomy, intrathecal baclofen (HIC)
Ophthalmologist, audiologist, dentist	Squint, glasses, cortical visual impairment; hearing aids; caries, drooling-related problems
Psychologist, social worker, teacher, CBR worker	Behaviour, family support, disability registration, **inclusive or special needs education**, community inclusion, parent groups

UGANDA CONTEXT

At HC III/IV level the realistic team is the **clinician, a nurse/midwife and a CBR worker or VHT**, with referral to a regional physiotherapy department, Mulago, Mengo or CoRSU. **Teaching the mother** is the core intervention: daily positioning, stretching, feeding upright with thickened food, a simple home-made **corner seat or standing frame**, and play. Parent support groups (e.g. "Getting to Know Cerebral Palsy") reduce stigma and maternal depression.

B. Motor management: step by step

1. **Physiotherapy and positioning (everyone)**: daily stretching of hip adductors, hamstrings and calf muscles; **24-hour postural care** (supported side-lying or supine with legs apart at night, supported sitting by day, prone lying); standing frames to load bones and stretch muscles; mobility training appropriate to the GMFCS level.
2. **Orthoses and casting**: AFOs to hold the ankle at 90° and improve gait; **serial casting** for equinus; hip abduction wedges/braces; hand splints.
3. **Oral drugs for generalised spasticity or dystonia** (table below): used when tone causes pain, care difficulty (nappy changes, hygiene), sleep problems or limits function.
4. **Focal treatment**: **botulinum toxin A** injected into specific spastic muscles (calf, hamstrings, adductors, elbow flexors) with casting and physiotherapy; effect lasts 3–6 months. Specialist, limited availability in Uganda.
5. **Surgery**: tendon lengthening (Achilles, hamstrings, adductors), hip reconstruction (osteotomy), scoliosis correction; **selective dorsal rhizotomy** for selected spastic diplegic walkers; **intrathecal baclofen** pump for severe spasticity/dystonia (HIC).

C. Drugs for tone

Drug	Dose	Route	Frequency	Max dose	Notes
Baclofen (GABA-B agonist)	Start **0.3 mg/kg/day** (or 2.5–5 mg/dose); increase every 3 days to usual **0.75–2 mg/kg/day**	Oral	Divided **3–4 times daily**	**40 mg/day <8 y; 60 mg/day ≥8 y** (adult 80 mg/day)	Sedation, hypotonia (trunk/head control may worsen), constipation, may lower seizure threshold. **Never stop abruptly**: withdrawal causes seizures, hallucinations, rebound spasticity, hyperthermia. Better for spinal than cerebral spasticity but worth a trial
Diazepam (GABA-A)	**0.1–0.8 mg/kg/day** (often 0.1–0.2 mg/kg at night)	Oral	Divided **6–8-hourly**, or at night	Adult 10 mg/dose	Short-term, painful spasms, night-time, peri-operative. Sedation, drooling, tolerance, dependence
Trihexyphenidyl (anticholinergic)	Start about **1 mg/day** (0.05–0.1 mg/kg/day); increase weekly	Oral	Divided 2–3 times daily	Specialist titration (verify local protocol)	**Dyskinetic/dystonic CP** (NICE NG62: with specialist input). Also reduces drooling. Dry mouth, constipation, urinary retention, blurred vision, behaviour change
Levodopa (with carbidopa)	Trial about 1 mg/kg/day levodopa, increasing slowly	Oral	Divided 2–3 times daily	Specialist	**Always trial** in unexplained dystonia (dopa-responsive dystonia); may help some dyskinetic CP
Tizanidine (α2 agonist)	Specialist	Oral			Rarely available in Uganda; hypotension, liver toxicity

Drug	Dose	Route	Frequency	Max dose	Notes
Botulinum toxin A	Units/kg per muscle by product (e.g. Botox total roughly ≤10–12 units/kg per session)	IM into target muscles	Every 3–6 months	Product-specific	Focal spasticity; must be combined with physio/casting; specialist only

DRUG DOSING

Baclofen withdrawal is dangerous. If a child on baclofen is admitted nil by mouth (e.g. pneumonia, surgery), **continue it via NG tube** or taper; never just stop it.

D. Epilepsy

About **1 in 3** children with CP has epilepsy (range 15–60%; highest in quadriplegia and hemiplegia, and with intellectual disability).

Drug (UCG 2023 choice)	Starting dose	Maintenance	Route / frequency	Notes
Carbamazepine (focal and generalised tonic–clonic; >2 y, or <2 y)	**5 mg/kg/day**	**10–20 mg/kg/day**	Oral, 2–3 divided doses	Rash (stop if severe), ataxia, diplopia; avoid in myoclonic/absence seizures
Sodium valproate (generalised, myoclonic, absence; >2 y)	10–15 mg/kg/day	**20–30 mg/kg/day** (max 60)	Oral, 2 divided doses	Weight gain, tremor, hepatotoxicity (risk highest under 2 y with metabolic disease); **teratogenic** (counsel adolescent girls)
Phenobarbital (<2 y per UCG; cheap and widely available)	3–5 mg/kg/day	3–5 mg/kg/day	Oral, once daily at night	**UCG: avoid phenobarbital and phenytoin in children with intellectual disability or behaviour problems** (worsens behaviour and cognition)

Acute seizures and status: as in the febrile convulsions topic (diazepam 0.5 mg/kg PR or buccal midazolam 0.3 mg/kg; phenobarbital 15–20 mg/kg; phenytoin 15–20 mg/kg). Give families with frequent prolonged seizures a written **rescue plan**.

E. Nutrition, feeding and drooling

1. **Assess** at every visit: weight, height/knee height, MUAC, Z-scores; feeding time (>30 minutes per meal is a warning), coughing/choking, chest infections.
2. **Feed safely**: **upright (90°)** seated with head slightly flexed, never lying flat; small spoonfuls; **thickened** feeds (thick porridge, mashed matooke, groundnut paste) rather than thin liquids; pace the feed; oral hygiene.
3. **Energy-dense diet**: add oil, groundnut/simsim paste, milk, eggs; RUTF/F-100 if SAM; vitamin and mineral supplements.
4. **NG tube** for short-term support or during acute illness; **gastrostomy** (PEG) for long-term unsafe swallowing or failure to gain weight (available at Mulago and some private/mission hospitals; families need counselling).
5. **GORD**: positioning, small frequent feeds, omeprazole (about 1 mg/kg/day) if oesophagitis suspected.
6. **Constipation** (very common): fluids, fibre, **lactulose** (about 0.5–1 mL/kg/dose once or twice daily) or polyethylene glycol; treat impaction.
7. **Bone health**: vitamin D 400 IU/day (more if deficient), calcium-rich diet, sunlight, weight-bearing; beware fractures with low trauma.

8. **Drooling**: positioning, oromotor therapy; anticholinergics (glycopyrronium, trihexyphenidyl, hyoscine patch: NICE NG62), specialist botulinum toxin to salivary glands.

F. Other associated problems

Problem	Action
Vision (strabismus, refractive error, cortical visual impairment)	Ophthalmology early; glasses; patching
Hearing	OAE/ABR early (especially after kernicterus, meningitis, CMV); hearing aids
Communication	SLT; augmentative and alternative communication (picture boards, gestures, devices)
Pain	Look for hips, constipation, reflux, dental, fractures, menstrual pain; regular paracetamol 15 mg/kg 6-hourly (max 60 mg/kg/day) while the cause is treated
Sleep	Treat pain, reflux, seizures; sleep hygiene; melatonin (specialist)
Respiratory	Treat aspiration pneumonia (UCG antibiotics; cover anaerobes if severe), immunise (PCV, measles), chest physiotherapy, safe feeding
Dental	Brushing, fluoride, dental review (caries, gum hyperplasia on phenytoin)
Behaviour, learning, school	Psychology; inclusive education; special schools; disability registration
Family and carers	Screen mother for depression, link to parent groups and CBR, social work, respite; address stigma and myths

G. Follow-up schedule (practical)

When	What
Every visit (1–3-monthly when young or unwell)	Growth and MUAC, feeding, seizures and drug doses (re-calculate by weight), chest, pain, sleep, development, family coping
6–12-monthly	GMFCS review, contractures, **hip X-ray** (GMFCS III–V, or II with concerns), spine, orthoses fit
Once early, then as needed	Vision and hearing assessments, MRI/CT (if not done), HIV status
Transition (adolescence)	Puberty, menstrual care, sexuality and safeguarding, vocational training, adult services

Complications

Musculoskeletal	Medical	Psychosocial
Contractures (equinus, knee flexion, hip adduction/flexion)	Aspiration pneumonia (leading cause of death)	Stigma, exclusion from school
Hip subluxation/dislocation (up to 60–90% in GMFCS V)	Malnutrition, micronutrient deficiency	Maternal depression, family breakdown
Scoliosis	Epilepsy, status epilepticus	Abuse and neglect (higher risk)
Osteopenia and low-trauma fractures	GORD, constipation, urinary infections	Poverty from caring costs
Pressure sores	Dental caries, drooling, skin breakdown	Limited access to devices and therapy

Ward-round & exam pearls

Model presentation

> "Peter is a 4-year-old boy from Kayunga weighing 12.0 kg (WAZ −2.6), height 96 cm (HAZ −1.8), head circumference 46 cm (below the 3rd centile), MUAC 13.0 cm, with known spastic quadriplegic cerebral palsy, GMFCS level V, attributed to severe birth asphyxia after a prolonged home labour (did not cry at birth, fitted on day 1, admitted to a neonatal unit for 10 days). He is HIV-negative, fully immunised per the UNEPI card (including MR at 9 and 18 months) and fed thin millet porridge by bottle lying down. He has epilepsy on carbamazepine 50 mg twice daily (8 mg/kg/day) with about 2 seizures a month. He presents with 3 days of cough and fever and choking during feeds. On examination: T 38.6°C, RR 52, SpO_2 89% in air, crackles at the right base; increased tone in all limbs with brisk reflexes, scissoring, bilateral equinus, restricted hip abduction to 20° on the right; no head control. **Impression: aspiration pneumonia in a child with spastic quadriplegic CP, GMFCS V, with moderate underweight, microcephaly, suboptimally controlled epilepsy, unsafe feeding and a right hip at risk.** Plan: oxygen, IV antibiotics per UCG, NG feeds while acutely ill, increase carbamazepine towards 10–15 mg/kg/day, SLT feeding assessment and caregiver training on upright thickened feeds, nutrition plan, hip X-ray, physiotherapy referral, link the mother to a parent support group."

Viva questions (with answers)

1. **Define CP.** Permanent disorder of movement and posture due to a non-progressive lesion of the developing brain, often with epilepsy, sensory, cognitive, communication and behaviour problems.
2. **Why is it called non-progressive if the child gets contractures?** The brain lesion is static; secondary musculoskeletal problems develop with growth.
3. **Name the motor types and their lesions.** Spastic (pyramidal: cortex/white matter), dyskinetic (basal ganglia/thalamus), ataxic (cerebellum), mixed.
4. **Preterm baby → which type?** Spastic diplegia (PVL).
5. **What does kernicterus cause?** Dyskinetic CP, sensorineural deafness, upward-gaze palsy, dental enamel dysplasia, intellectual function often relatively preserved.
6. **What is GMFCS and why does it matter?** Five-level classification of gross motor function; predicts mobility, hip displacement, nutrition needs and survival.
7. **What would make you doubt the diagnosis?** Regression, progressive course, no risk factors, family history, normal MRI, peripheral signs.
8. **Drugs for spasticity?** Baclofen, diazepam, botulinum toxin A (focal), tizanidine; intrathecal baclofen in HIC. Dystonia: trihexyphenidyl, levodopa trial.

OSCE tips

- In an **examination station**, observe first (posture, movements, aids, orthoses), then tone, power, reflexes, plantar responses, clonus, primitive reflexes (in infants), joints (hips!), spine, gait. Finish by **classifying**: "spastic diplegia, GMFCS II" and listing what else you would assess (growth, vision, hearing, speech, feeding, seizures).
- In a **counselling station** (breaking the news): private setting, both parents, ask what they know, use simple words, explain that CP is not contagious or caused by witchcraft, focus on what the child **can** do and on practical steps (therapy, feeding, school), and arrange follow-up.

Mnemonics & memory aids

- **The "F-words" for childhood disability** (Rosenbaum and Gorter): **F**unction, **F**amily, **F**itness, **F**un, **F**riends, **F**uture. Set goals in each.
- **Associated problems "PLEASE SEE FOOD"**: **P**ain, **L**earning disability, **E**pilepsy, **A**spiration, **S**leep, **E**yes; **S**peech, **E**ars, **E**limination (constipation, bladder); **F**eeding, **O**steopenia, **O**rthopaedic (hips, spine), **D**rooling.
- **Topography by timing**: "**P**reterm = **P**araplegic-ish (diplegia, legs), **T**erm = **T**otal (quadriplegia) or **T**witchy (dyskinetic), **S**troke = **S**ide (hemiplegia)."
- **Kernicterus tetrad "DDUE"**: **D**yskinesia, **D**eafness, **U**pward-gaze palsy, **E**namel dysplasia.

High-yield summary

Numbers & doses to know

Item	Value
Prevalence	**2–3.5/1000** in HICs; about **3/1000** in Uganda (Iganga-Mayuge)
Spastic / dyskinetic / ataxic	**~70–80% / 10–15% / <5%**
Epilepsy in CP	**15–60%** (about 1 in 3)
Intellectual disability	**30–50%**
Lesion timing for CP	Fetal life to about **2 years**
NICE NG62 warning signs	Not sitting by **8 months**, not walking by **18 months**, hand preference before **1 year** (corrected age)
Primitive reflexes	Moro, ATNR, grasp normally gone by **4–6 months**; persistence suggests CP
Walking prognosis	Sits by **2 years** → likely to walk; not sitting by about 4 years → unlikely
Hip migration percentage	**>30%** at risk; **>40%** displaced → orthopaedics
Baclofen	Start **0.3 mg/kg/day** ÷ 3–4; usual **0.75–2 mg/kg/day**; max **40 mg/day (<8 y)**, **60 mg/day (≥8 y)**
Diazepam (spasticity)	**0.1–0.8 mg/kg/day** ÷ 6–8-hourly
Carbamazepine	**5 mg/kg/day → 10–20 mg/kg/day** (UCG 2023)
Sodium valproate	**10–15 → 20–30 mg/kg/day**
Phenobarbital maintenance	**3–5 mg/kg/day**
Paracetamol	**15 mg/kg 6-hourly**, max 60 mg/kg/day
Feeding warning	Meal time **>30 minutes**, coughing with feeds, recurrent chest infections

Classic exam traps

- **"The child lost the ability to sit, so the CP is getting worse."** Wrong: CP is non-progressive; regression means another diagnosis (HIV encephalopathy, metabolic, neurodegenerative).
- **"All CP is caused by birth asphyxia."** Wrong: most CP in HICs is prenatal; in Uganda infection, jaundice, prematurity and asphyxia all matter.
- **"Spastic diplegia = only the legs are affected."** Wrong: the arms are mildly involved.
- **"Toe-walking boy with delayed walking = CP."** Check CK and Gowers sign: Duchenne.

- **"Stop baclofen while nil by mouth."** Wrong: abrupt withdrawal causes seizures and hallucinations; give it by NG.
- **"Phenobarbital is ideal for a child with CP and behaviour problems."** Wrong: UCG advises avoiding phenobarbital and phenytoin in children with intellectual disability or behaviour problems.
- **"He is just thin because of CP."** Wrong: undernutrition is treatable and is a major cause of death.
- **"Feed lying down so he does not fall."** Wrong: feed upright with thickened textures to prevent aspiration.

Questions seniors ask

How would you classify this child's CP? By motor type (spastic, dyskinetic, ataxic, mixed), topography (hemi-, di-, quadriplegia), function (GMFCS, MACS, CFCS) and associated impairments, plus the likely cause and timing.

What makes you sure this is CP and not a progressive disorder? A clear risk factor (e.g. birth asphyxia, kernicterus, meningitis), steady slow gains with no regression, a static neurological picture, and, where available, a consistent MRI. Any regression, no risk factors, or a family history needs a neurology referral.

What are the main causes of death in severe CP, and how do you prevent them? Aspiration pneumonia and malnutrition. Safe upright feeding with thickened textures, SLT assessment, nutritional support (NG/gastrostomy), immunisation, early treatment of chest infections, seizure control.

Why do you X-ray the hips of a child who cannot walk? Spastic adductors and hip flexors plus lack of weight-bearing lead to progressive hip migration and painful dislocation; surveillance allows early soft-tissue surgery before dislocation.

How does baclofen work and what are its side effects? GABA-B agonist reducing spinal reflex excitability; sedation, hypotonia, constipation, lower seizure threshold, and a dangerous withdrawal syndrome if stopped suddenly.

What can a HC IV do for this child? Classify and document, treat acute illness and seizures, measure growth and manage malnutrition, teach positioning, stretching and safe feeding, test for HIV, refill antiepileptics, link to a CBR worker and refer for physiotherapy, imaging, hearing and vision.

What do you tell parents who ask whether their child will walk? It depends on the GMFCS level and on sitting: if the child sits alone by 2 years, walking (perhaps with aids) is likely; GMFCS IV–V children usually need wheelchairs. Focus on goals the child can achieve and on comfort, communication and participation.

SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define cerebral palsy.	A group of **permanent** disorders of **movement and posture**, causing activity limitation, due to a **non-progressive** disturbance in the **developing** fetal or infant brain; often with epilepsy, sensory, cognitive, communication and behaviour problems and secondary musculoskeletal problems (Rosenbaum 2007).
2	Up to what age can a brain injury cause CP?	Fetal life to about **2 years** of age (the developing brain).
3	Proportions of spastic, dyskinetic and ataxic CP?	Spastic **~70–80%**, dyskinetic **~10–15%**, ataxic **<5%**; mixed forms are common.
4	Which CP pattern follows periventricular leukomalacia in a preterm baby, and why?	**Spastic diplegia**: corticospinal fibres to the **legs** run closest to the ventricles in the periventricular white matter.
5	Which brain areas are damaged by acute profound asphyxia at term, and what CP results?	**Basal ganglia and thalamus → dyskinetic** CP (or mixed). Prolonged partial asphyxia → watershed cortex → spastic quadriplegia.
6	Tetrad of kernicterus sequelae?	**Dyskinetic CP**, **sensorineural deafness**, **upward-gaze palsy**, **dental enamel dysplasia** ('DDUE').
7	Commonest cause of spastic hemiplegia?	**Perinatal arterial stroke** (usually MCA territory). Postnatal: sickle cell stroke, head injury, meningitis.
8	GMFCS levels I–V (age 6–12)?	**I** walks without limitations; **II** walks with limitations; **III** walks with a hand-held mobility device; **IV** self-mobility limited, may use powered mobility; **V** transported in a manual wheelchair.
9	What does the GMFCS level predict?	Future mobility, **hip displacement**, scoliosis, feeding/nutrition needs and survival. It is fairly stable after age 2.
10	NICE NG62 motor warning signs of CP (corrected age)?	Not **sitting by 8 months**, not **walking by 18 months**, **hand preference before 1 year**; also abnormal tone, persistent fisting, toe-walking, feeding difficulties.
11	Red flags that a child's 'CP' may be another disorder?	**No risk factors**, **loss of skills**, **family history** of progressive neurological disease, **new focal signs**; also normal MRI, peripheral signs (absent reflexes, fasciculation), abnormal head growth.
12	Prevalence of CP in Uganda (Iganga-Mayuge population study)?	About **3 per 1000 children**, with more severe impairment, malnutrition, epilepsy and postnatal causes than in HIC registers.
13	Main causes of CP in Uganda?	**Birth asphyxia/HIE**, **prematurity**, **neonatal jaundice (kernicterus)**, **neonatal sepsis/meningitis**, hypoglycaemia, **cerebral malaria**, **bacterial meningitis**, head injury, sickle cell stroke, congenital infections.
14	Why must every child with developmental delay be tested for HIV in Uganda?	**HIV encephalopathy** causes delay/regression, acquired microcephaly and spasticity, and improves with **ART** (UCG 2023 recommends testing).
15	How common is epilepsy in CP, and which types carry most risk?	**15–60%** (about 1 in 3); highest in **spastic quadriplegia** and hemiplegia, and with intellectual disability.
16	Leading causes of death in severe CP?	**Aspiration pneumonia** and **malnutrition** (also status epilepticus).
17	Oral baclofen dosing in a child?	Start **0.3 mg/kg/day** in 3–4 divided doses, increase every 3 days to **0.75–2 mg/kg/day**; max **40 mg/day (<8 y)**, **60 mg/day (≥8 y)**.

#	Question	Answer
18	Main danger of baclofen?	**Abrupt withdrawal**: seizures, hallucinations, rebound spasticity, hyperthermia. Never stop suddenly; give via NG if nil by mouth.
19	Oral diazepam dose for spasticity in children?	**0.1–0.8 mg/kg/day** divided 6–8-hourly (often a night-time dose); short-term use because of sedation, drooling and tolerance.
20	Drug used for dystonic/dyskinetic CP, and the trial you must not forget?	**Trihexyphenidyl** (specialist input, NICE NG62); always try **levodopa** in unexplained dystonia to catch **dopa-responsive dystonia**.
21	Treatment for focal spasticity (e.g. calf causing equinus)?	**Botulinum toxin A** injection plus **serial casting**/AFOs and physiotherapy; effect lasts 3–6 months.
22	UCG 2023 antiepileptic choice by age for generalised tonic–clonic seizures?	**<2 years: phenobarbital or carbamazepine**; **>2 years: carbamazepine or valproate**. Avoid phenobarbital and phenytoin with intellectual disability/behaviour problems.
23	Carbamazepine dosing in children (UCG 2023)?	Start **5 mg/kg/day**, maintenance **10–20 mg/kg/day** in divided doses.
24	Hip surveillance in CP: test and thresholds?	**AP pelvis X-ray** for Reimers **migration percentage**: **>30% at risk**, **>40% displaced** → orthopaedic referral. Start about age 2, especially GMFCS III–V.
25	Safe feeding advice for a child with CP and poor swallowing?	Feed **upright (90°)** with head slightly flexed, **thickened** textures, small spoonfuls, slow pace; energy-dense food; SLT assessment; NG/gastrostomy if unsafe or poor growth.
26	The 'F-words' in childhood disability?	**Function, Family, Fitness, Fun, Friends, Future** (Rosenbaum and Gorter): set goals in each area.
27	Walking prognosis rule of thumb in CP?	A child who **sits independently by 2 years** will very likely walk (with or without aids); not sitting by about **4 years** → unlikely to walk.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 2-year-old boy born at 29 weeks (birthweight 1.1 kg) is not yet walking. He sits independently and uses his hands to play, but his legs are stiff with brisk knee jerks, ankle clonus and scissoring on vertical suspension. Arms show only mild clumsiness. Which is the MOST likely diagnosis?

- A Spastic hemiplegic CP from a perinatal MCA stroke
- B Dyskinetic CP from kernicterus
- C Spastic diplegic CP from periventricular leukomalacia
- D Ataxic CP from a cerebellar malformation
- E Duchenne muscular dystrophy

2. A 3-year-old girl was diagnosed with 'cerebral palsy' at 18 months because of delayed walking. Her mother now says she used to say 10 words and feed herself, but over 6 months she has stopped talking, has lost weight and her head circumference has fallen from the 50th to below the 3rd centile. Her mother died of an unknown illness last year. Which is the MOST appropriate next step?

- A HIV test (rapid test per the national algorithm) and refer for ART if positive
- B Reassure: CP often appears to worsen as the child grows
- C Start baclofen to improve her motor function
- D Order a hip X-ray for hip surveillance
- E Refer to physiotherapy only

3. A 6-year-old boy with CP walks indoors with a rollator walker and uses a wheelchair pushed by his mother for trips to the market and school. Which GMFCS level BEST describes him?

- A Level I
- B Level II
- C Level IV
- D Level V
- E Level III

4. A 5-year-old girl with dyskinetic CP has writhing movements of her arms that worsen when she tries to reach, intermittent arching, and drooling. She also does not respond to her name, and her eyes do not look upward. Her mother recalls she was 'very yellow' in the first week of life and had a blood exchange. What should be arranged as a PRIORITY for her development? Advanced

- A Brain MRI before anything else
- B Formal hearing assessment (OAE/ABR) and audiology
- C Hip X-ray every 6 months
- D Serum creatine kinase
- E EEG

5. A 4-year-old boy (12 kg, WAZ −2.6) with spastic quadriplegic CP, GMFCS V, is admitted with fever, RR 54 and right basal crackles. His mother says meals take an hour and he coughs and splutters with every feed of thin porridge given by bottle while lying down. After treating the pneumonia, which intervention will MOST reduce his risk of death?

A Starting oral baclofen
B Weekly physiotherapy for his contractures
C Hip X-ray surveillance
D Safe feeding plan: upright position, thickened energy-dense feeds, SLT assessment and NG/gastrostomy if still unsafe
E Changing his antiepileptic to phenobarbital

6. A 7-year-old (20 kg) with spastic diplegia takes baclofen 10 mg three times daily. He is admitted with gastroenteritis and kept nil by mouth on IV fluids; his baclofen is stopped. On day 2 he becomes agitated, hallucinates, has a temperature of 39.5°C, increased stiffness and a generalised seizure. Which is the MOST likely cause? Advanced

A Baclofen withdrawal
B Febrile convulsion
C Hypernatraemic dehydration
D Bacterial meningitis from the gut
E Neuroleptic malignant syndrome

7. A 9-year-old boy (25 kg) with spastic hemiplegic CP has new focal seizures with secondary generalisation twice a month. He has mild intellectual disability and is often aggressive at school. Using UCG 2023, which is the MOST appropriate first antiepileptic?

A Phenobarbital 75 mg once daily
B Phenytoin 125 mg twice daily
C Carbamazepine 5 mg/kg/day, increasing to 10–20 mg/kg/day in divided doses
D Ethosuximide
E No treatment until he has status epilepticus

8. A 2-year-old boy is referred as 'cerebral palsy' because he walked at 19 months, walks on his toes and falls often. Pregnancy and birth were normal. He has large firm calves, rises from the floor by pushing on his thighs, and has reduced knee jerks. What is the MOST appropriate next investigation?

A MRI brain
B EEG
C Hip X-ray
D TSH
E Serum creatine kinase

9. A 5-year-old girl with GMFCS IV spastic quadriplegia cries during nappy changes. Hip abduction is limited to 15° on the left. An AP pelvis X-ray shows a left migration percentage of 48% and right 18%. What is the MOST appropriate action? Advanced

A Reassure: migration up to 60% is normal in CP
B Refer to orthopaedics for surgical management of the left hip, with analgesia and positioning meanwhile
C Start oral baclofen and repeat the X-ray in 2 years
D Apply a full leg cast for 6 weeks
E Prescribe calcium only

10. A 10-month-old boy born at home after a 2-day labour did not cry at birth and fitted on day 1. Now he cannot sit, keeps his hands fisted with thumbs adducted, has a persistent asymmetric tonic neck reflex and stiff legs that cross when held upright. What is the BEST next step at a HC IV?

A Tell the mother he will catch up and review at 2 years
B Start phenobarbital to prevent seizures
C Refer immediately for selective dorsal rhizotomy
D Explain the likely diagnosis of CP, test for HIV, measure growth, teach positioning/stretching/feeding and refer for physiotherapy and further assessment
E Request a muscle biopsy

11. A 6-year-old girl with 'dyskinetic CP' was born at term with a normal birth and never had jaundice. Her leg dystonia is mild in the morning but by evening she can barely walk. MRI brain is normal. What is the MOST appropriate next step? Advanced

A Start trihexyphenidyl and accept the diagnosis
B Botulinum toxin to both calves
C Bilateral Achilles tendon lengthening
D Phenobarbital at night
E A trial of levodopa (with carbidopa)

12. A 4-year-old boy (15 kg) with spastic diplegia, GMFCS II, has painful night-time spasms and stiffness that make sitting at school difficult. Which is the MOST appropriate oral starting regimen?

A Baclofen about 1.5 mg three times daily (0.3 mg/kg/day), increasing every 3 days
B Baclofen 20 mg three times daily
C Diazepam 10 mg four times daily
D Phenytoin 5 mg/kg/day
E Carbamazepine 10 mg/kg/day

13. Which pairing of cause and CP pattern is CORRECT?

A Periventricular leukomalacia → ataxic CP
B Kernicterus → spastic hemiplegia
C Left middle cerebral artery perinatal stroke → right spastic hemiplegia
D Cerebellar hypoplasia → dyskinetic CP
E Acute profound asphyxia at term → spastic diplegia

14. The mother of a 3-year-old with newly diagnosed CP says her relatives believe the child is cursed, and her husband has left. She asks whether the child should stay home from nursery school. What is the MOST appropriate response?

A Advise keeping him at home until he can walk
B Explain that CP is caused by an injury to the developing brain, is not a curse or contagious, encourage inclusive schooling, and link her to a parent support group, CBR worker and social worker
C Refer her to a traditional healer to satisfy the relatives
D Tell her that nothing can be done so she should focus on her other children
E Advise residential institutional care

15. Which statement about cerebral palsy is TRUE?

- **A** Most children with CP have severe intellectual disability
- **B** The brain lesion slowly progresses throughout childhood
- **C** Most CP in high-income countries is due to events during labour
- **D** Spastic CP is the most common motor type
- **E** Ataxic CP is the most common type after preterm birth

Answers & explanations

1. C · Spastic diplegic CP from periventricular leukomalacia

C is right: a **very preterm** baby with **legs much more affected than arms**, UMN signs and scissoring has **spastic diplegia from PVL** (leg fibres lie nearest the ventricles). A is wrong: hemiplegia affects one side, arm usually more than leg. B is wrong: kernicterus causes involuntary movements and fluctuating tone, not pure spastic legs. D is wrong: ataxic CP gives hypotonia, wide-based gait and tremor, not spasticity. E is wrong: Duchenne causes proximal weakness with reduced reflexes and normal early milestones, not spasticity in a preterm infant.

2. A · HIV test (rapid test per the national algorithm) and refer for ART if positive

A is right: **regression** plus **acquired microcephaly**, weight loss and a mother who died of an unknown illness strongly suggests **HIV encephalopathy**; UCG 2023 says test every child with developmental delay for HIV. ART can halt progression. B is wrong: CP is non-progressive; loss of skills means another diagnosis. C is wrong: baclofen treats spasticity, not the cause. D is wrong: hip surveillance does not address a progressive, treatable disease. E is wrong: physiotherapy alone misses a treatable cause.

3. E · Level III

E is right: walking with a **hand-held mobility device** (walker or crutches) indoors and using a wheelchair for longer distances is **GMFCS III**. A is wrong: level I walks without limitations. B is wrong: level II walks without aids but with limitations (long distances, uneven ground). C is wrong: level IV has limited self-mobility and may use powered mobility; they do not walk functionally with a walker. D is wrong: level V children are transported in a wheelchair and have limited head and trunk control.

4. B · Formal hearing assessment (OAE/ABR) and audiology

B is right: dyskinetic CP, upward-gaze palsy and neonatal jaundice requiring exchange point to **kernicterus**, which causes **sensorineural deafness**; not responding to her name may be hearing loss rather than cognitive impairment. Early hearing aids and a communication plan matter most. A is wrong: MRI may confirm globus pallidus injury but does not change her immediate developmental care. C is wrong: hip surveillance is relevant but not the most urgent need here. D is wrong: CK is for suspected muscular dystrophy. E is wrong: there is no history of seizures.

5. D · Safe feeding plan: upright position, thickened energy-dense feeds, SLT assessment and NG/gastrostomy if still unsafe

D is right: **aspiration pneumonia and malnutrition** are the leading causes of death in severe CP; upright feeding with thickened textures, nutrition support and tube feeding when swallowing is unsafe address both. A is wrong: baclofen treats tone and may worsen trunk control and secretions. B is wrong: physiotherapy is important but does not prevent aspiration deaths. C is wrong: hip surveillance prevents painful dislocation, not death. E is wrong: there is no reason to change, and phenobarbital is sedating and worsens behaviour and cognition.

6. A · Baclofen withdrawal

A is right: **abrupt baclofen withdrawal** causes agitation, hallucinations, hyperthermia, rebound spasticity and seizures; restart it (by NG) and taper only if needed. B is wrong: he is 7, outside the febrile convulsion age range, and hallucinations are not part of it. C is wrong: he is on IV fluids and hypernatraemia would not explain rebound spasticity and hallucinations. D is wrong: gastroenteritis does not cause meningitis, and the timing fits withdrawal. E is wrong: he has received no antipsychotic drug.

7. C · Carbamazepine 5 mg/kg/day, increasing to 10–20 mg/kg/day in divided doses

C is right: UCG 2023 recommends **carbamazepine** (or valproate) for children >2 years, starting **5 mg/kg/day** and increasing to **10–20 mg/kg/day**; it suits focal seizures. A and B are wrong: UCG advises **avoiding phenobarbital and phenytoin** in children with intellectual disability or behaviour problems. D is wrong: ethosuximide treats absence seizures only. E is wrong: recurrent seizures need treatment; waiting for status is dangerous.

8. E · Serum creatine kinase

E is right: a boy with **no CP risk factors**, toe-walking, **calf pseudohypertrophy**, **Gowers sign** and **reduced reflexes** probably has **Duchenne muscular dystrophy**; CK is markedly raised. A is wrong: the signs are of muscle disease, not a brain lesion. B is wrong: there are no seizures. C is wrong: hip X-ray is for hip surveillance in established CP. D is wrong: hypothyroidism causes hypotonia and delay but not Gowers sign with pseudohypertrophy.

9. B · Refer to orthopaedics for surgical management of the left hip, with analgesia and positioning meanwhile

B is right: a migration percentage **>40% = displacement**; with pain and limited abduction she needs **orthopaedic referral** (adductor release/osteotomy) plus pain relief and positioning. A is wrong: >30% is already at risk. C is wrong: baclofen does not reverse hip displacement, and 2 years is far too long to wait. D is wrong: casting does not treat hip displacement. E is wrong: calcium does not address the mechanical problem.

10. D · Explain the likely diagnosis of CP, test for HIV, measure growth, teach positioning/stretching/feeding and refer for physiotherapy and further assessment

D is right: risk factors (asphyxia, neonatal seizures) plus early signs (not sitting, fisting, persistent ATNR beyond 6 months, scissoring) indicate likely **CP**; early intervention, HIV testing (UCG), nutrition and referral for MDT care are the priorities. A is wrong: waiting loses the period of greatest brain plasticity. B is wrong: antiepileptics are not given prophylactically. C is wrong: rhizotomy is a selective specialist procedure for older spastic diplegic walkers, not first management. E is wrong: the signs are of an upper motor neurone lesion, not muscle disease.

11. E · A trial of levodopa (with carbidopa)

E is right: **no risk factors**, a **normal MRI** and **diurnal variation** suggest **dopa-responsive dystonia**, which responds dramatically to low-dose levodopa. A is wrong: accepting CP misses a treatable mimic. B is wrong: focal treatment does not address the cause. C is wrong: surgery is inappropriate for a treatable neurochemical disorder. D is wrong: phenobarbital does not treat dystonia and causes sedation.

12. A · Baclofen about 1.5 mg three times daily (0.3 mg/kg/day), increasing every 3 days

A is right: baclofen starts at about **0.3 mg/kg/day** (15 × 0.3 = 4.5 mg/day ≈ 1.5 mg three times daily) and is titrated every 3 days towards 0.75–2 mg/kg/day, max 40 mg/day under 8 years. B is wrong: 60 mg/day exceeds the maximum for a child under 8 (40 mg/day) and starts far too high. C is wrong: 40 mg/day of diazepam (about 2.7 mg/kg/day) is a dangerous overdose; the range is 0.1–0.8 mg/kg/day. D and E are wrong: phenytoin and carbamazepine are antiepileptics, not antispasticity drugs.

13. C · Left middle cerebral artery perinatal stroke → right spastic hemiplegia

C is right: a **left MCA** infarct causes a **right-sided** spastic hemiplegia (arm > leg). A is wrong: PVL causes spastic diplegia. B is wrong: kernicterus causes dyskinetic CP. D is wrong: cerebellar lesions cause ataxic CP. E is wrong: acute profound asphyxia at term injures the basal ganglia/thalamus (dyskinetic) and prolonged partial asphyxia causes quadriplegia; diplegia is typical of preterm PVL.

14. B · Explain that CP is caused by an injury to the developing brain, is not a curse or contagious, encourage inclusive schooling, and link her to a parent support group, CBR worker and social worker

B is right: addressing myths, encouraging **participation and inclusive education**, and linking the family to **parent groups, CBR and social support** is family-centred care that reduces stigma and maternal depression. A is wrong: exclusion harms development and participation. C is wrong: it reinforces harmful beliefs and delays effective care. D is wrong: many interventions improve function, comfort and participation. E is wrong: institutional care is not first-line; children do best with supported families.

15. D · Spastic CP is the most common motor type

D is right: **spastic** CP accounts for about 70–80% of cases. A is wrong: intellectual disability affects about 30–50%, and many (especially diplegia and dyskinetic CP) have normal intelligence. B is wrong: the lesion is non-progressive; only secondary problems change. C is wrong: most CP in HICs has prenatal origins; intrapartum asphyxia explains a minority. E is wrong: spastic diplegia is typical after preterm birth.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: The baby who did not cry at birth

Nansubuga brings her 11-month-old son Joel to the Mengo paediatric clinic because 'he cannot sit and his legs are stiff'. He was born at home after a long labour, did not cry for 'a long time' and fitted on the first day. Weight 7.4 kg (WAZ −2.0), length 70 cm (HAZ −1.4), HC 42.5 cm (below the 3rd centile), MUAC 12.8 cm.

1. What further history do you need?

> Milestones by domain and **any regression**; antenatal illness, HIV/syphilis status; labour details, resuscitation, neonatal admission, **jaundice**, hypoglycaemia, sepsis; postnatal malaria/meningitis/head injury; seizures now; vision/hearing; **feeding** (time, coughing, chest infections); immunisation; HIV status; family history; social support and beliefs.

2. Examination shows head lag, fisted hands with adducted thumbs, persistent ATNR and Moro, brisk reflexes with ankle clonus and scissoring on vertical suspension. What is your assessment?

> Likely **spastic quadriplegic CP** due to **birth asphyxia (HIE)**, with **microcephaly**, borderline **underweight** and global developmental delay. Persistent primitive reflexes beyond 6 months and scissoring support CP.

3. What investigations are appropriate?

> **HIV test** (UCG 2023), glucose and Hb if unwell, sickle test if indicated; **cranial ultrasound/CT or MRI** where available (cause, hydrocephalus); hearing (OAE/ABR) and vision assessment; EEG only if seizures; TSH if hypotonia or other features.

4. How do you break the news?

> Private room, both parents if possible, ask what they know, use simple words: 'part of Joel's brain was injured around birth; this affects his movement and it will not get worse'. It is not a curse, not contagious, not her fault. Focus on what he can do and what will help. Allow questions; arrange follow-up and written information.

5. What is your management plan?

> **MDT referral**: physiotherapy (positioning, stretching, supported sitting), OT, **SLT feeding assessment**; **nutrition**: upright feeding, thickened energy-dense food, monitor weight monthly; hearing and vision services; seizure plan if fits; immunisation catch-up; link to **CBR worker and a parent group** (e.g. Getting to Know Cerebral Palsy); social work.

6. What will you monitor at follow-up?

> Growth and MUAC, feeding safety and chest infections, seizures, tone and contractures, **GMFCS** classification from about age 2, **hip X-ray** from about 2 years (high risk), vision, hearing, communication, maternal mental health, school inclusion later.

Take-away: Early diagnosis, nutrition, feeding safety and family support are the most powerful interventions for a Ugandan child with CP.

Case 2: Is it really cerebral palsy?

Akello, a 3-year-old girl from Gulu, is referred with 'CP and fits'. She walked at 14 months and spoke in short phrases at 2 years. Over the last 8 months she has stopped speaking, walks on her toes with stiff legs, and has lost weight. Her mother is on ART but Akello has never been tested.

1. What is the key feature that argues against CP?

> **Regression**: she achieved walking and phrases normally and then lost skills. CP is **non-progressive**; loss of skills means a progressive or acquired disorder.

2. What is the most likely diagnosis, and what other causes would you consider?

> **HIV encephalopathy** (HIV-exposed, never tested, weight loss, spasticity, regression). Others: TB meningitis sequelae, neurodegenerative/metabolic disease, brain tumour, Rett syndrome (girl with regression, hand-wringing), subacute sclerosing panencephalitis after measles.

3. What examination findings would you look for?

> **Head circumference** trend (acquired microcephaly), growth/MUAC (wasting), lymphadenopathy, oral thrush, hepatosplenomegaly, chronic otitis, signs of TB; neurology: spasticity, reflexes, cranial nerves, fundoscopy; hand stereotypies (Rett).

4. Which investigations will you order first?

> **HIV rapid test** (child >18 months: antibody algorithm), CD4 and viral load if positive; **CrAg** if CD4 low; CXR and Xpert Ultra if TB suspected; CT/MRI brain (atrophy, basal ganglia calcification); FBC; EEG for fits.

5. She is HIV-positive with a CD4 of 9%. What is the management?

> Start **ART** urgently per the current Uganda consolidated HIV guidelines (after excluding cryptococcal and TB meningitis), **co-trimoxazole prophylaxis**, TB screening and preventive therapy as indicated, nutrition rehabilitation, antiepileptic (carbamazepine; check ART interactions), physiotherapy and family counselling; adherence support.

6. What can the family expect?

> ART often **halts progression** and allows some recovery of skills, though deficits may persist. Regular follow-up of viral load, growth and development; rehabilitation and schooling support; test siblings.

Take-away: Regression is never CP; in Uganda test every child with developmental delay for HIV.

Case 3: The yellow baby who became a floppy, writhing toddler

Mugisha, a 2.5-year-old boy (11 kg), is brought to a regional referral hospital because he cannot sit unsupported and 'his arms and legs move on their own'. He was born at a HC III on day 3 of a public holiday weekend and became very yellow on day 2; he reached hospital on day 6 and had an exchange transfusion. He smiles and laughs appropriately but does not turn to his name.

1. What type of CP is this, and what is the cause?

> **Dyskinetic (choreoathetoid/dystonic) CP** from **kernicterus** (bilirubin encephalopathy) after severe neonatal hyperbilirubinaemia treated late.

2. What other features of kernicterus will you look for?

> **Sensorineural hearing loss** (auditory neuropathy), **upward-gaze palsy**, **dental enamel dysplasia** of primary teeth; intellect often relatively preserved (his social smile and laughter are reassuring).

3. What are the immediate priorities in assessment?

> **Hearing assessment** (OAE may be normal in auditory neuropathy; ABR needed), growth and feeding (dyskinetic children have high energy needs), oromotor function and drooling, seizures, hip examination, communication assessment, HIV status, immunisation.

4. Outline his management.

> Physiotherapy (positioning, supported seating), OT (adapted seating, hand use), **SLT and augmentative communication**, hearing aids/audiology, energy-dense nutrition, **trihexyphenidyl** for dystonia with specialist input (start about 1 mg/day and titrate), drooling management, family support and inclusive education.

5. How could this have been prevented?

> Check every newborn for **jaundice** before discharge and at follow-up visits, teach mothers to recognise yellow skin and poor feeding, measure bilirubin (transcutaneous or serum) and plot against hour-specific thresholds (AAP 2022), early **phototherapy** at HC IV and above, prompt referral for exchange transfusion, and screening for risk factors (prematurity, G6PD deficiency, ABO/Rh incompatibility, sepsis).

Take-away: Kernicterus causes dyskinetic CP and deafness; it is entirely preventable with early jaundice detection and phototherapy.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.